# IMMERSIONS INTO A PRESENTED STACK COME FROM R-INVARIANT SUBSPACES OF THE PRESENTATION

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 1895-1910 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 1912-1933, the complete author proof (20 body lines).

*Editorial note.* Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 23 of queue_stacks-properties.json (pb=768, stmt 1895-1910, proof 1912-1933).

*Editorial note.* stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma lemma-representable-in-terms-presentations: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 770–787. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

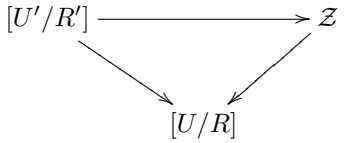
**Lemma 0.1.** *Let $(U, R, s, t, c)$ be a smooth groupoid in algebraic spaces. Let $i : \mathcal{Z} \to [U/R]$ be an immersion. Then there exists an $R$-invariant locally closed subspace $Z \subset U$ and a presentation $[Z/R_Z] \to \mathcal{Z}$ where $R_Z$ is the restriction of $R$ to $Z$ such that*

$$\begin{array}{ccc} [Z/R_Z] & \longrightarrow & \mathcal{Z} \\ & \searrow \quad \swarrow_{i} & \\ & [U/R] & \end{array}$$

*is 2-commutative. If $i$ is a closed (resp. open) immersion then $Z$ is a closed (resp. open) subspace of $U$.*

*Date*: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* By Lemma lemma-representable-in-terms-presentations we get a commutative diagram




where $U' = \mathcal{Z} \times_{[U/R]} U$ and $R' = \mathcal{Z} \times_{[U/R]} R$. Since $\mathcal{Z} \to [U/R]$ is an immersion we see that $U' \to U$ is an immersion of algebraic spaces. Let $Z \subset U$ be the locally closed subspace such that $U' \to U$ factors through $Z$ and induces an isomorphism $U' \to Z$. It is clear from the construction of $R'$ that $R' = U' \times_{U,t} R = R \times_{s,U} U'$. This implies that $Z \cong U'$ is $R$-invariant and that the image of $R' \to R$ identifies $R'$ with the restriction $R_Z = s^{-1}(Z) = t^{-1}(Z)$ of $R$ to $Z$. Hence the lemma holds. $\square$